

EE-102: Electric Circuit Analysis

LECTURE NO 3: *Series DC Circuits*

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Organization of Lecture

1. Introduction
2. Series Resistors
3. Series Circuits
4. Power distribution in a Series Circuit
5. Voltage Sources in Series
6. Voltage Division in a Series Circuit
7. Notation

1. Introduction

- Two types of current are readily available to the consumer today. One is *direct current*(dc),in which ideally the flow of charge (current) does not change in magnitude (or direction) with time. The other is *sinusoidal alternating current*(ac),in which the flow of charge is continually changing in magnitude (and direction) with time.
- The battery in Fig. 1 next slide, by virtue of the potential difference between its terminals, has the ability to cause charge to flow through the simple circuit. As long as the battery is connected in the circuit and maintains its terminal characteristics, the current (dc) through the circuit will not change in magnitude (value) or direction.
- If we consider the wire to be an ideal conductor (that is, having no opposition to flow),the potential difference V across the resistor equals the applied voltage of the battery: $V(\text{volts}) = E(\text{volts})$.

1. Introduction(continued...)

- The current is limited only by the resistor R . The higher the resistance, less is the current, and conversely, as determined by Ohm's law.
- By convention, the direction of **conventional current flow** ($I_{\text{conventional}}$) as shown in Fig. 1 is opposite to that of **electron flow** (I_{electron}). Also, the uniform flow of charge dictates that the direct current I be the same everywhere in the circuit.

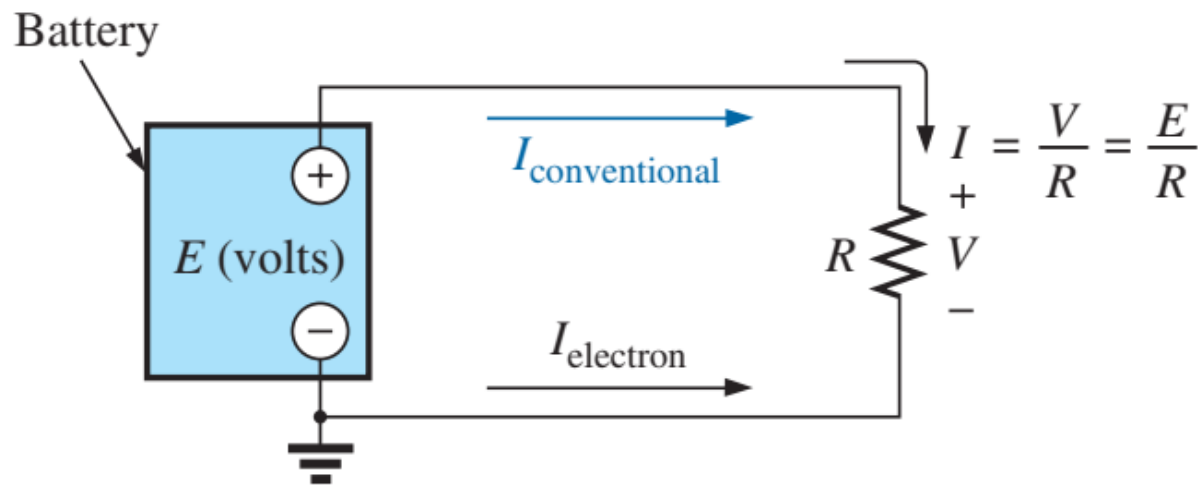
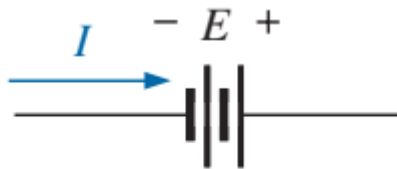


Figure 1

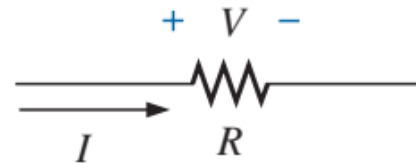
1. Introduction (continued...)

- By following the direction of conventional flow, we notice that there is a rise in potential across the battery (- to +) and a drop in potential across the resistor (+ to -).
- For single-voltage-source dc circuits, conventional flow always passes from a low potential to a high potential when passing through a voltage source, as shown in Fig. 2. However, conventional flow always passes from a high to a low potential when passing through a resistor for any number of voltage sources in the same circuit, as shown in Fig. 3.



For all one-voltage-source dc circuits

Figure 2



For any combination of voltage sources in the same dc circuit

Figure 3

2. Series Resistors

- Every fixed resistor has only two terminals to connect in a configuration - it is therefore referred to as a **two-terminal device**.
- In Fig. 4, one terminal of resistor R_2 is connected to resistor R_1 on one side, and the remaining terminal is connected to resistor R_3 on the other side, resulting in one, and only one, connection between adjoining resistors.
- When connected in this manner, the resistors have established a series connection. If three elements were connected to the same point, as shown in Fig. 5, there would not be a series connection between resistors R_1 and R_2 .

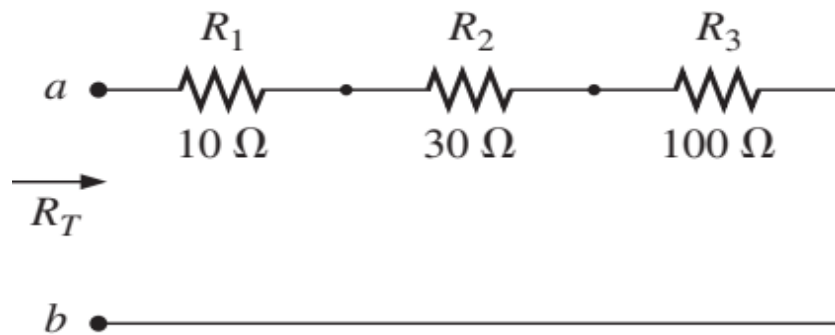


FIG. 4

Series connection of resistors.

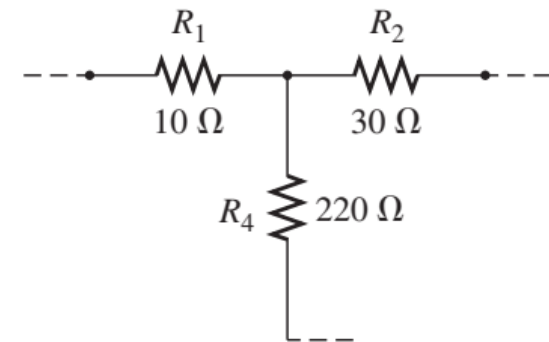


FIG. 5

Configuration in which none of the resistors are in series.

2. Series Resistors (continued...)

- *The total resistance of a series configuration is the sum of the resistance levels.*

In equation form for any number (N) of resistors,

$$R_T = R_1 + R_2 + R_3 + R_4 + \cdots + R_N \quad (1)$$

- *The more resistors we add in series, the greater is the resistance, no matter what their value.*

2. Series Resistors (continued...)

EXAMPLE 1 Determine the total resistance of the series connection in Fig. 6. Note that all the resistors appearing in this network are standard values.

Solution: Note in Fig. 6 that even though resistor R_3 is on the vertical and resistor R_4 returns at the bottom to terminal b , all the resistors are in series since there are only two resistor leads at each connection point.

Applying Eq. (1) gives

$$R_T = R_1 + R_2 + R_3 + R_4$$

$$R_T = 20\ \Omega + 220\ \Omega + 1.2\ \text{k}\Omega + 5.6\ \text{k}\Omega$$

and

$$R_T = 7040\ \Omega = \mathbf{7.04\ \text{k}\Omega}$$

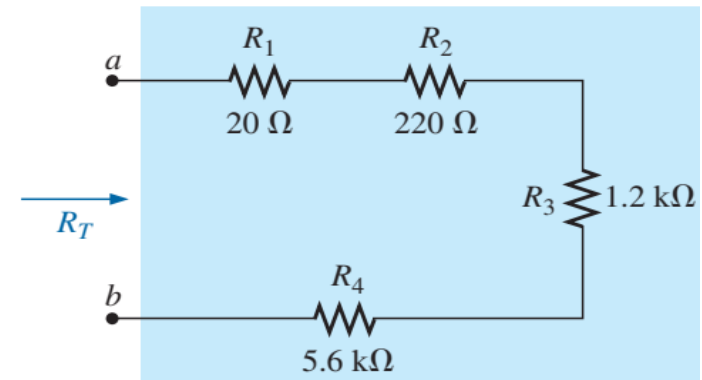


FIG. 6

Series connection of resistors for Example 1.

2. Series Resistors (continued...)

- For the special case where resistors are the same value, Eq. (1) can be modified as follows:

$$R_T = NR$$

(2)

Where N is the number of resistors in series of value R .

2. Series Resistors (continued...)

EXAMPLE 2 Find the total resistance of the series resistors in Fig. 7. Again, recognize $3.3 \text{ k}\Omega$ as a standard value.

Solution: Again, don't be concerned about the change in configuration. Neighboring resistors are connected only at one point, satisfying the definition of series elements.

$$\begin{aligned}\text{Eq. (2):} \quad R_T &= NR \\ &= (4)(3.3 \text{ k}\Omega) = \mathbf{13.2 \text{ k}\Omega}\end{aligned}$$

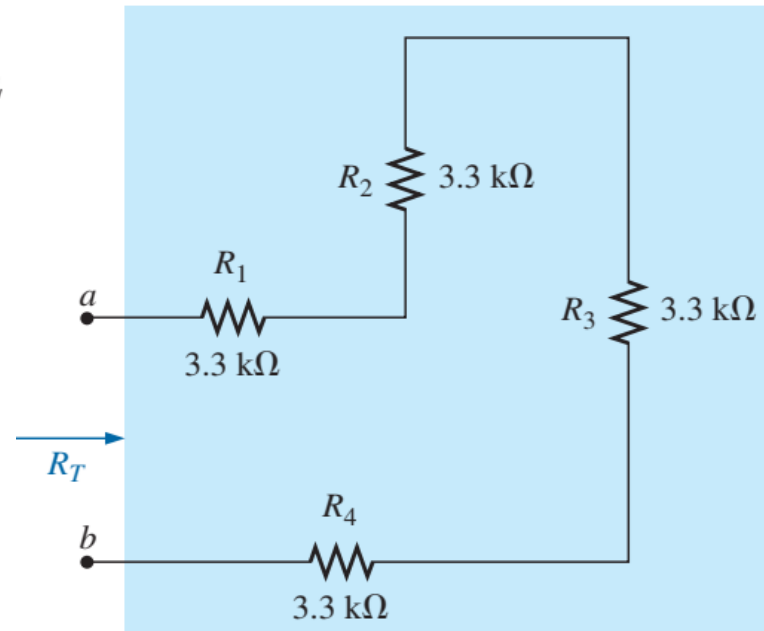


FIG. 7

Series connection of four resistors of the same value (Example 2).

2. Series Resistors (continued...)

- *The total resistance of resistors in series is unaffected by the order in which they are connected.*
- The result is that the total resistance in Fig. 8(a) is the same as in Fig. 8(b).

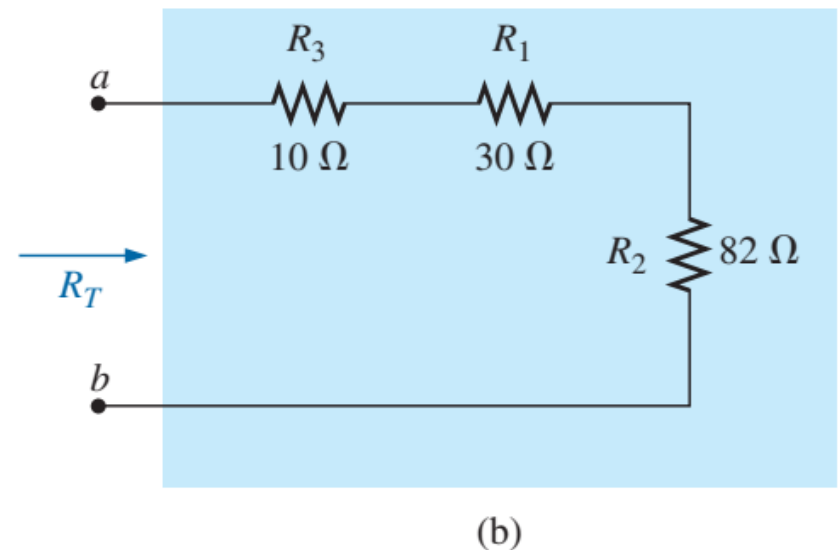
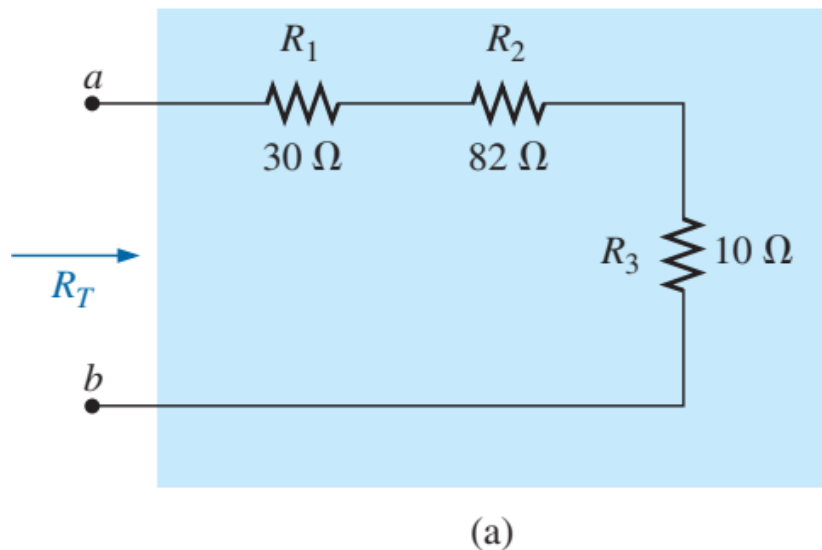


FIG. 8

Two series combinations of the same elements with the same total resistance.

3. Series Circuits

- If we now take an 8.4 V dc supply and connect it in series with the series resistors in Fig. 4, we have the series circuit in Fig. 9.
- *A circuit is any combination of elements that will result in a continuous flow of charge, or current, through the configuration.*

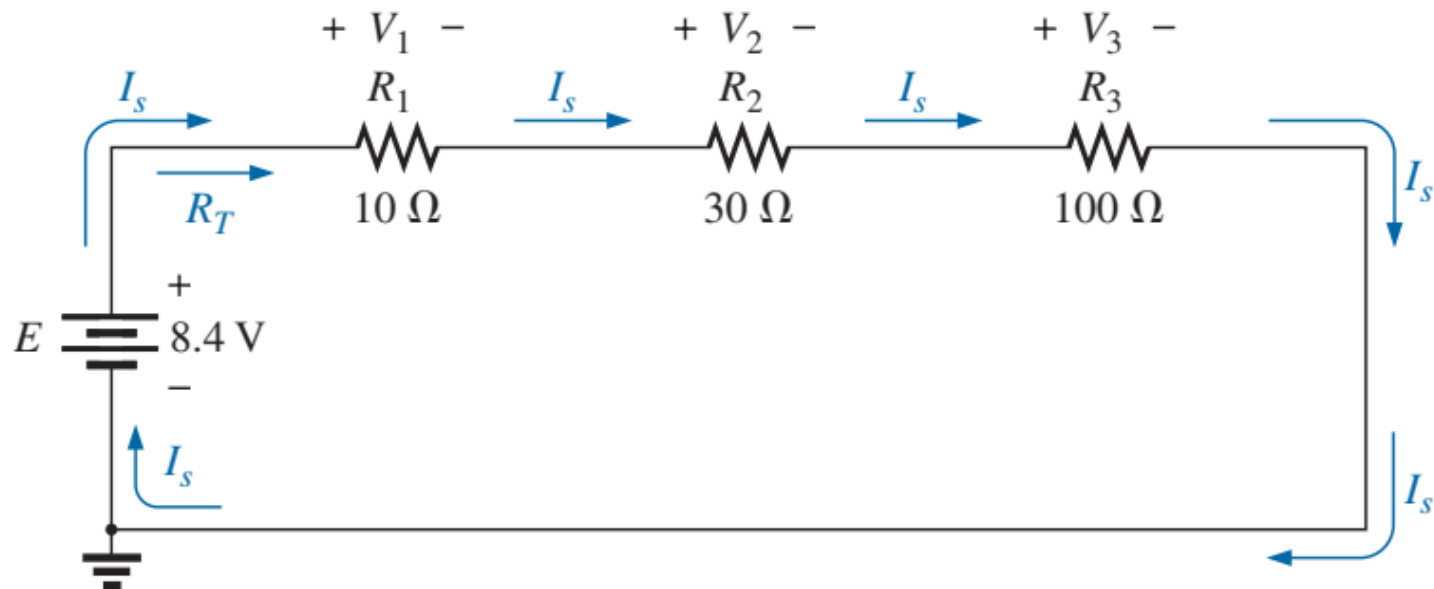


Figure 9

3. Series Circuits (continued...)

- The manner in which the supply is connected determines the direction of the resulting conventional current. For series dc circuits:
 - *The direction of conventional current in a series dc circuit is such that it leaves the positive terminal of the supply and returns to the negative terminal, as shown in Fig. 9.*
- One of the most important concepts to remember when analysing series circuits and defining elements that are in series is:
 - *The current is the same at every point in a series circuit.*
- Now that we have a complete circuit and current has been established, the level of current and the voltage across each resistor should be determined. To do this, return to Ohm's law and replace the resistance in the equation by the total resistance of the circuit. That is,

$$E = I_s R_T \longrightarrow I_s = \frac{E}{R_T} \quad (3)$$

3. Series Circuits (continued...)

- *The polarity of the voltage across a resistor is determined by the direction of the current.*
- Current entering a resistor creates a drop in voltage with the polarity indicated in Fig. 10(a). Reverse the direction of the current, and the polarity will reverse as shown in Fig. 10(b). Change the orientation of the resistor, and the same rules apply as shown in Fig. 10(c). Applying the above to the circuit in Fig. 9 will result in the polarities appearing in that figure.

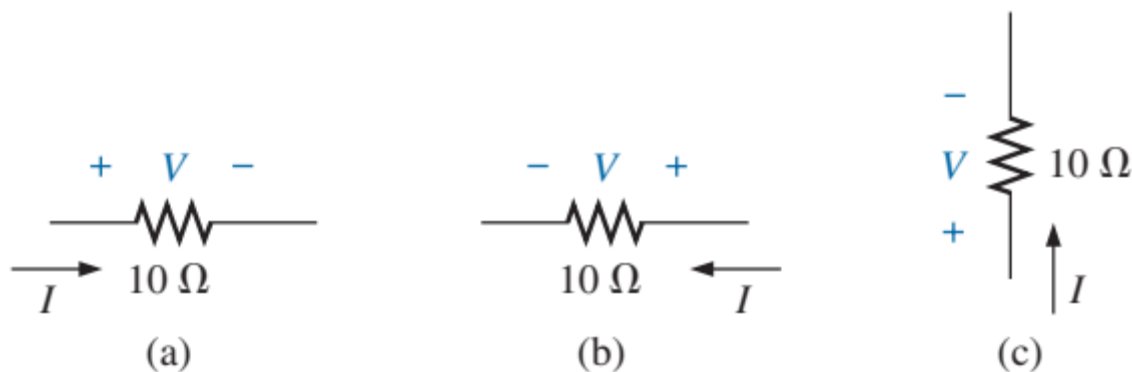


Figure 10: Inserting the polarities across a resistor as determined by the direction of the current.

3. Series Circuits (continued...)

Example 4 For the series circuit in fig. 11:

- Find the total resistance R_T .
- Calculate the resulting source current I_s .
- Determine the voltage across each resistor.

Solutions:

a. $R_T = R_1 + R_2 + R_3$
 $= 2\ \Omega + 1\ \Omega + 5\ \Omega$
 $R_T = \mathbf{8\ \Omega}$

b. $I_s = \frac{E}{R_T} = \frac{20\ \text{V}}{8\ \Omega} = \mathbf{2.5\ \text{A}}$

c. $V_1 = I_1 R_1 = I_s R_1 = (2.5\ \text{A})(2\ \Omega) = \mathbf{5\ \text{V}}$

$$V_2 = I_2 R_2 = I_s R_2 = (2.5\ \text{A})(1\ \Omega) = \mathbf{2.5\ \text{V}}$$

$$V_3 = I_3 R_3 = I_s R_3 = (2.5\ \text{A})(5\ \Omega) = \mathbf{12.5\ \text{V}}$$

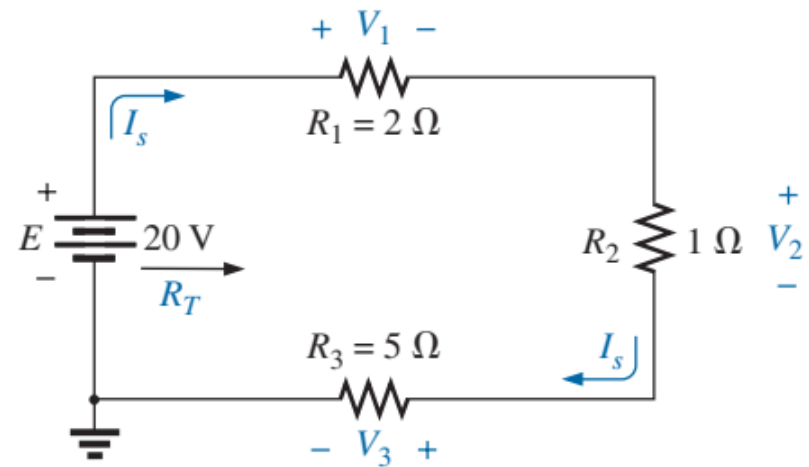


Figure 11: Series circuit to be investigated in Example 4.

3. Series Circuits (continued...)

Homework!

Example 5 For the series circuit in fig. 12:

- Find the total resistance R_T .
- Determine the source current I_s and indicate its direction on the circuit.
- Find the voltage across resistor R_2 and indicate its polarity on the circuit.

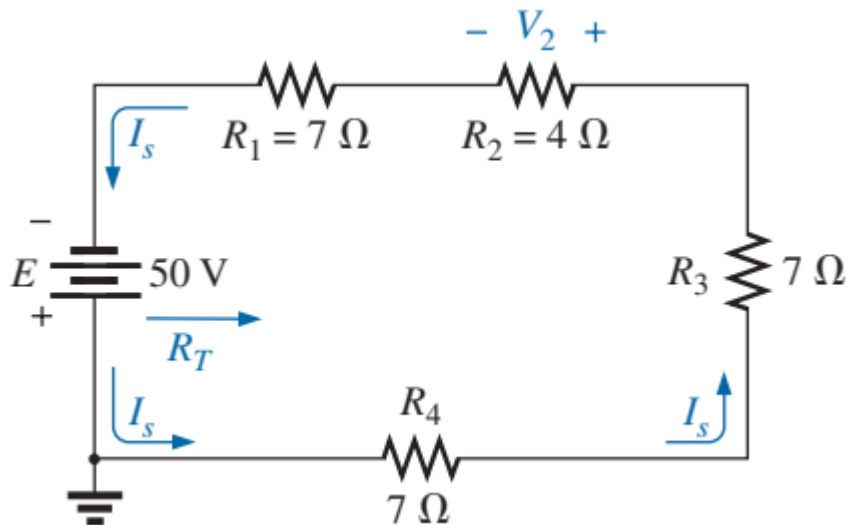


Figure 12: Series circuit to be investigated in Example 5.

3. Series Circuits (continued...)

Example 6 Given R_T and I_3 , calculate R_1 and E for the circuit in fig. 13:

Solution: Since we are given the total resistance, it seems natural to first write the equation for the total resistance and then insert what we know:

$$R_T = R_1 + R_2 + R_3$$

We find that there is only one unknown, and it can be determined with some simple mathematical manipulations. That is,

$$12\text{ k}\Omega = R_1 + 4\text{ k}\Omega + 6\text{ k}\Omega = R_1 + 10\text{ k}\Omega$$

and $12\text{ k}\Omega - 10\text{ k}\Omega = R_1$

so that $R_1 = 2\text{ k}\Omega$

The dc voltage can be determined directly from Ohm's law:

$$E = I_s R_T = I_3 R_T = (6\text{ mA})(12\text{ k}\Omega) = 72\text{ V}$$

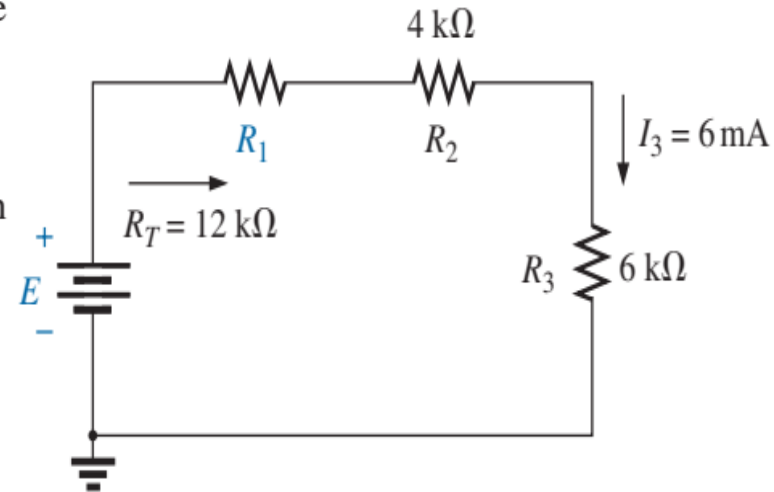


Figure 13: Series circuit to be investigated in Example 6.

4. Power distribution in a Series Circuit

- In any electrical system, the power applied will equal the power dissipated or absorbed. For any series circuit, such as that in Fig. 14.
- *The power applied by the dc supply must equal that dissipated by the resistive elements.*

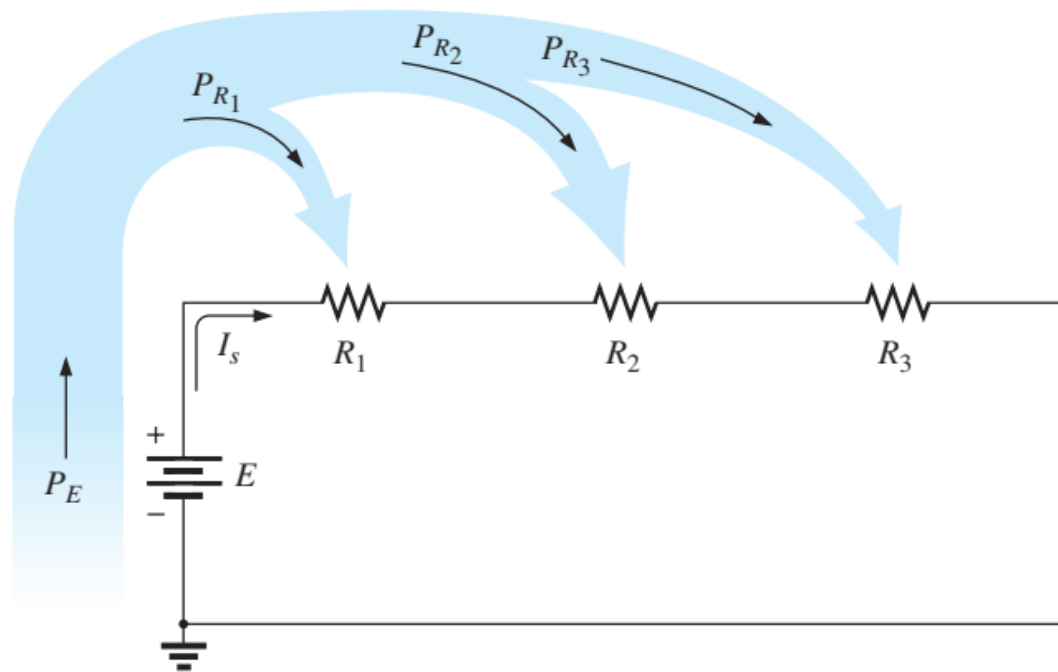


Figure 14: Power distribution in a series circuit.

4. Power distribution in a Series Circuit (continued...)

In equation form,

$$P_E = P_{R_1} + P_{R_2} + P_{R_3} \quad (5)$$

The power delivered by the supply can be determined using

$$P_E = EI_s \quad (\text{watts, W}) \quad (6)$$

The power dissipated by the resistive elements can be determined by any of the following forms (shown for resistor R_1 only):

$$P_1 = V_1 I_1 = I_1^2 R_1 = \frac{V_1^2}{R_1} \quad (\text{watts, W}) \quad (7)$$

in a series configuration, maximum power is delivered to the largest resistor.

4. Power distribution in a Series Circuit (continued...)

Example 7 For the circuit in fig. 15:

- Determine the total resistance R_T .
- Calculate the current I_s .
- Determine the voltage across each resistor.
- Find the power supplied by the battery.
- Determine the power dissipated by each resistor.
- Comment on whether the total power supplied equals the total power dissipated.

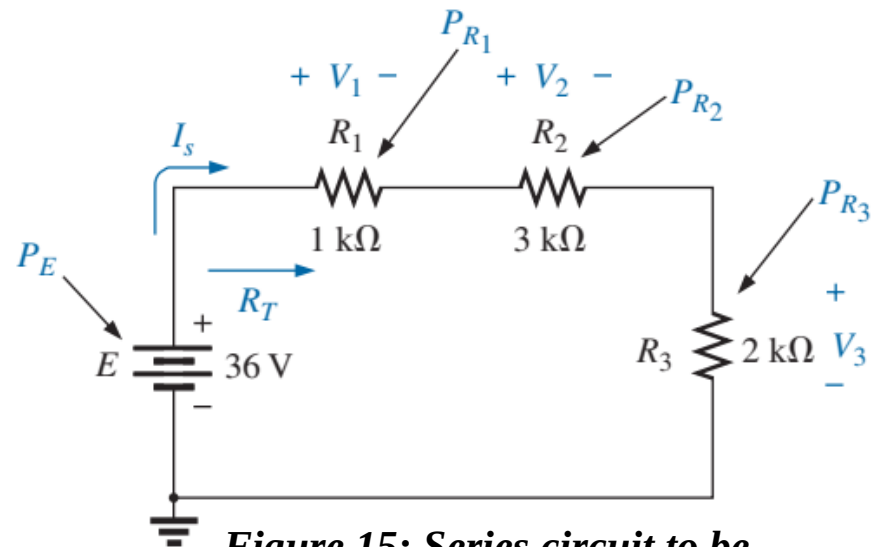


Figure 15: Series circuit to be investigated in Example 7.

4. Power distribution in a Series Circuit (continued...)

Example 7

Solutions:

$$\begin{aligned}\text{a. } R_T &= R_1 + R_2 + R_3 \\ &= 1 \text{ k}\Omega + 3 \text{ k}\Omega + 2 \text{ k}\Omega\end{aligned}$$

$$R_T = \mathbf{6 \text{ k}\Omega}$$

$$\text{b. } I_s = \frac{E}{R_T} = \frac{36 \text{ V}}{6 \text{ k}\Omega} = \mathbf{6 \text{ mA}}$$

$$\text{c. } V_1 = I_1 R_1 = I_s R_1 = (6 \text{ mA})(1 \text{ k}\Omega) = \mathbf{6 \text{ V}}$$

$$V_2 = I_2 R_2 = I_s R_2 = (6 \text{ mA})(3 \text{ k}\Omega) = \mathbf{18 \text{ V}}$$

$$V_3 = I_3 R_3 = I_s R_3 = (6 \text{ mA})(2 \text{ k}\Omega) = \mathbf{12 \text{ V}}$$

$$\text{d. } P_E = EI_s = (36 \text{ V})(6 \text{ mA}) = \mathbf{216 \text{ mW}}$$

$$\text{e. } P_1 = V_1 I_1 = (6 \text{ V})(6 \text{ mA}) = \mathbf{36 \text{ mW}}$$

$$P_2 = I_2^2 R_2 = (6 \text{ mA})^2 (3 \text{ k}\Omega) = \mathbf{108 \text{ mW}}$$

$$P_3 = \frac{V_3^2}{R_3} = \frac{(12 \text{ V})^2}{2 \text{ k}\Omega} = \mathbf{72 \text{ mW}}$$

$$\begin{aligned}\text{f. } P_E &= P_{R_1} + P_{R_2} + P_{R_3} \\ 216 \text{ mW} &= 36 \text{ mW} + 108 \text{ mW} + 72 \text{ mW} = \mathbf{216 \text{ mW}} \quad (\text{checks})\end{aligned}$$

5. Voltage Sources in Series

- Voltage sources can be connected in series, as shown in Fig. 16, to increase or decrease the total voltage applied to a system. The net voltage is determined by summing the sources with the same polarity and subtracting the total of the sources with the opposite polarity. The net polarity is the polarity of the larger sum.

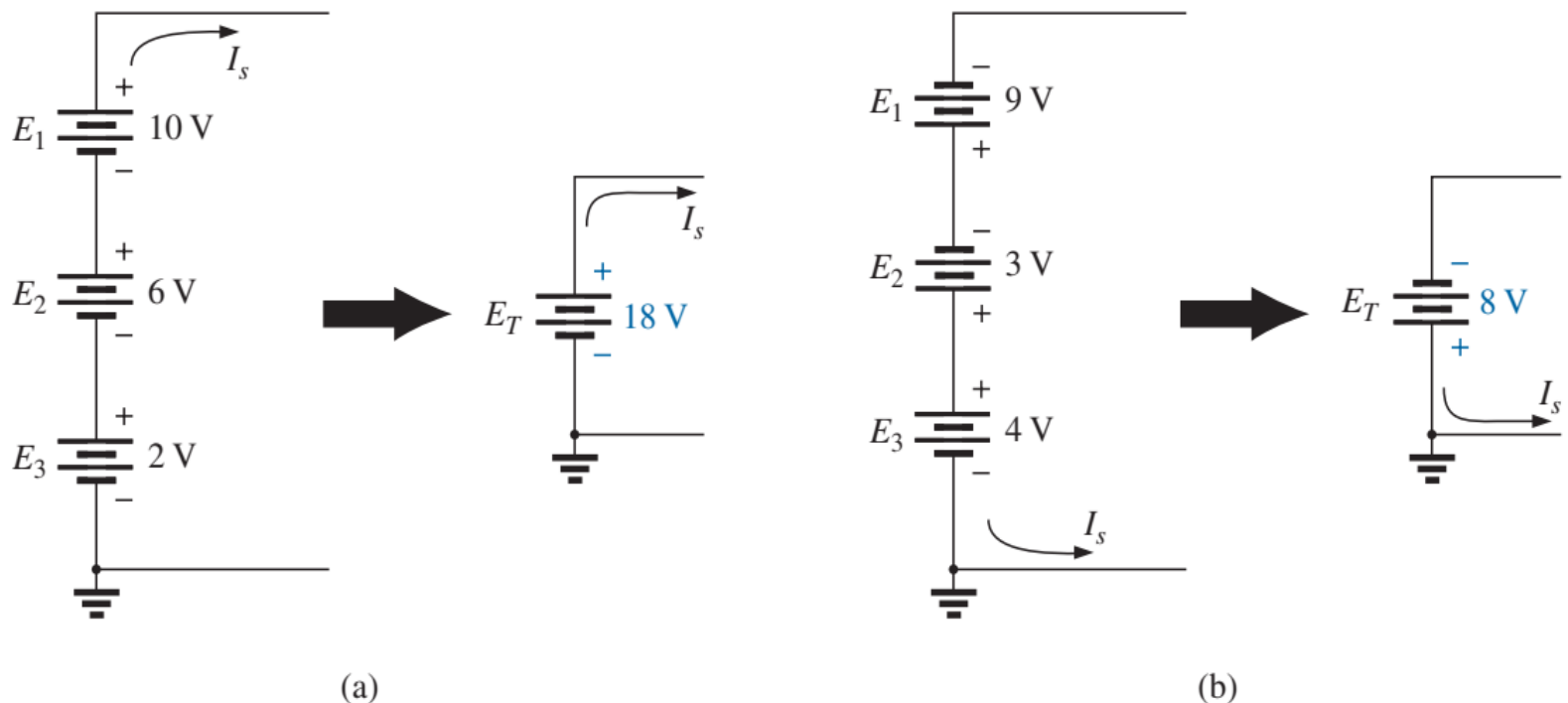


Figure 16: Reducing series dc voltage sources to a single source.

5. Voltage Sources in Series (continued...)

- In Fig. 16(a), for example, the sources are all “pressuring” current to follow a clockwise path, so the net voltage is

$$E_T = E_1 + E_2 + E_3 = 10 \text{ V} + 6 \text{ V} + 2 \text{ V} = \mathbf{18 \text{ V}}$$

- As shown in the figure. In Fig. 16(b), however, the 4 V source is “pressuring” current in the clockwise direction while the other two are trying to establish current in the counter clockwise direction. In this case, the applied voltage for a counter clockwise direction is greater than that for the clockwise direction. The result is the counter clockwise direction for the current as shown in Fig. 16(b). The net effect can be determined by finding the difference in applied voltage between those supplies “pressuring” current in one direction and the total in the other direction. In this case, with the polarity shown in the figure.

$$E_T = E_1 + E_2 - E_3 = 9 \text{ V} + 3 \text{ V} - 4 \text{ V} = \mathbf{8 \text{ V}}$$

5. Voltage Sources in Series (continued...)

Example 8 For the circuit in fig. 17, determine unknown voltage:

Solution: In this case, the unknown voltage is not across a single resistive element but between two arbitrary points in the circuit. Simply apply Kirchhoff's voltage law around a path, including the source or resistor R_3 . For the clockwise path, including the source, the resulting equation is the following:

$$+E - V_1 - V_x = 0$$

and
$$V_x = E - V_1 = 32 \text{ V} - 12 \text{ V} = \mathbf{20 \text{ V}}$$

For the clockwise path, including resistor R_3 , the following results:

$$+V_x - V_2 - V_3 = 0$$

and
$$\begin{aligned} V_x &= V_2 + V_3 \\ &= 6 \text{ V} + 14 \text{ V} \end{aligned}$$

with
$$V_x = \mathbf{20 \text{ V}}$$

providing exactly the same solution.

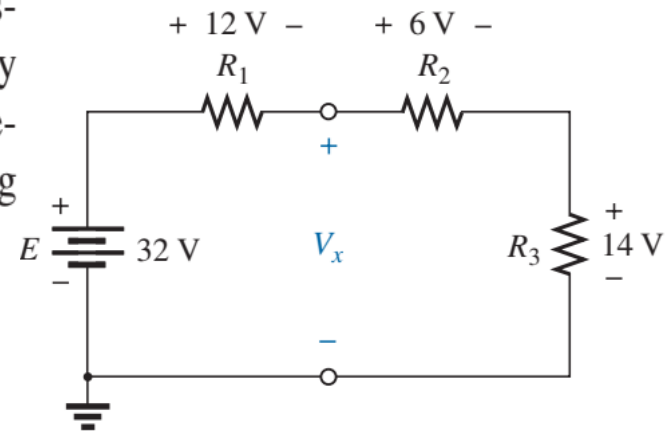


Figure 17: Series dc circuit to be analysed in Example 8.

5. Voltage Sources in Series (continued...)

Example 9 Determine the voltage V_x for the circuit in Fig. 18. Note that the polarity of V_x was not provided:

Solution: For cases where the polarity is not included, simply make an assumption about the polarity, and apply Kirchhoff's voltage law as before. If the result has a positive sign, the assumed polarity was correct. If the result has a minus sign, the **magnitude is correct**, but the assumed polarity must be reversed. In this case, if we assume point a to be positive and point b to be negative, an application of Kirchhoff's voltage law in the clockwise direction results in

$$-6\text{ V} - 14\text{ V} - V_x + 2\text{ V} = 0$$

and

$$V_x = -20\text{ V} + 2\text{ V}$$

so that

$$V_x = -18\text{ V}$$

Since the result is negative, we know that point a should be negative and point b should be positive, but the magnitude of 18 V is correct.

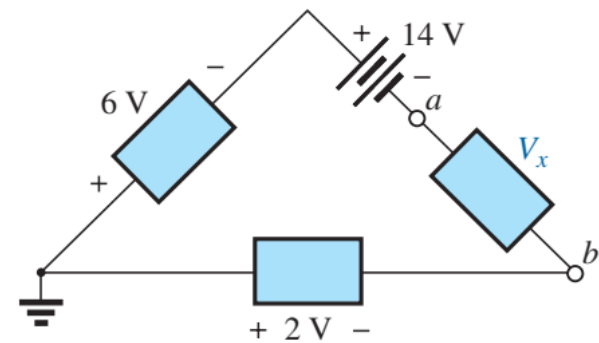


Figure 18: Series dc circuit to be analysed in Example 9.

6. Voltage Division in a Series Circuit

- *In a series resistive circuit, the larger the resistance, the more of the applied voltage it will capture.* In addition
- *The ratio of the voltages across series resistors will be the same as the ratio of their resistance levels.*
- In In Fig.19 next slide, all the voltages across the resistive elements are provided. The largest resistor of 6Ω captures the bulk of the applied voltage, while the smallest resistor, R_3 , has the least. In addition, note that since the resistance level of R_1 is six times that of R_3 , the voltage across R_1 is six times that of R_3 . The fact that the resistance level of R_2 is three times that of R_1 results in three times the voltage across R_2 . Finally, since R_1 is twice R_2 , the voltage across R_1 is twice that of R_2 . In general, therefore, the voltage across series resistors will have the same ratio as their resistance levels.

6. Voltage Division in a Series Circuit (continued...)

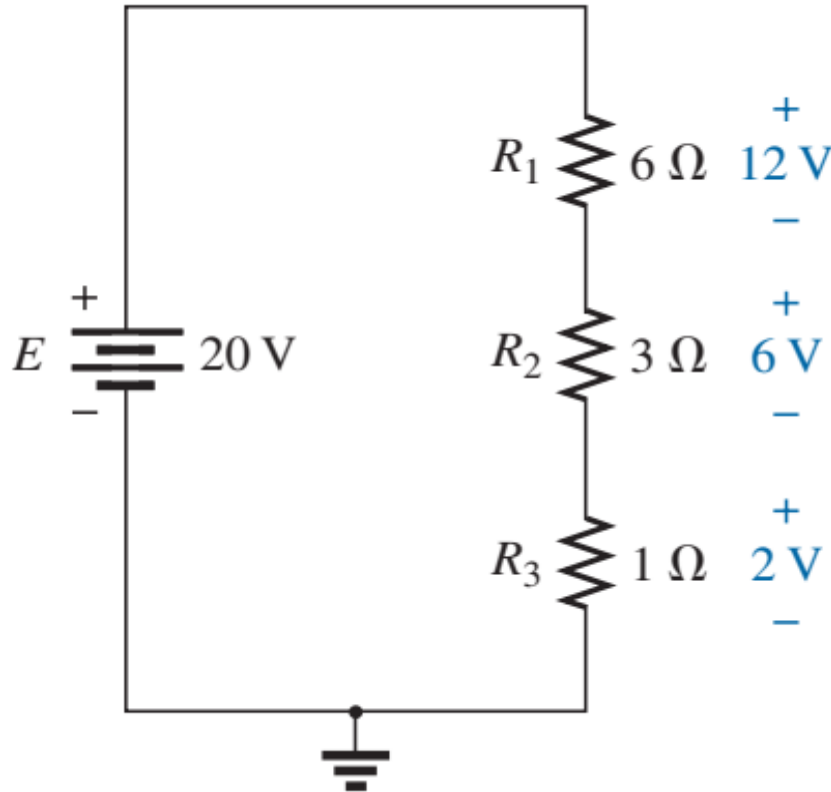


Figure 19: Revealing how the voltage will divide across series resistive elements.

6. Voltage Division in a Series Circuit (continued...)

Voltage Divider Rule (VDR)

- The voltage divider rule (VDR) permits the determination of the voltage across a series resistor without first having to determine the current of the circuit. The rule itself can be derived by analyzing the simple series circuit in Fig. 20 next slide.

First, determine the total resistance as follows:

$$R_T = R_1 + R_2$$

Then

$$I_s = I_1 = I_2 = \frac{E}{R_T}$$

Apply Ohm's law to each resistor:

$$V_1 = I_1 R_1 = \left(\frac{E}{R_T} \right) R_1 = R_1 \frac{E}{R_T}$$

$$V_2 = I_2 R_2 = \left(\frac{E}{R_T} \right) R_2 = R_2 \frac{E}{R_T}$$

The resulting format for V_1 and V_2 is

$$V_x = R_x \frac{E}{R_T}$$

(voltage divider rule)

6. Voltage Division in a Series Circuit (continued...)

Voltage Divider Rule (VDR)

- The VDR states that:
 - *The voltage across a resistor in a series circuit is equal to the value of that resistor times the total applied voltage divided by the total resistance of the series configuration.*

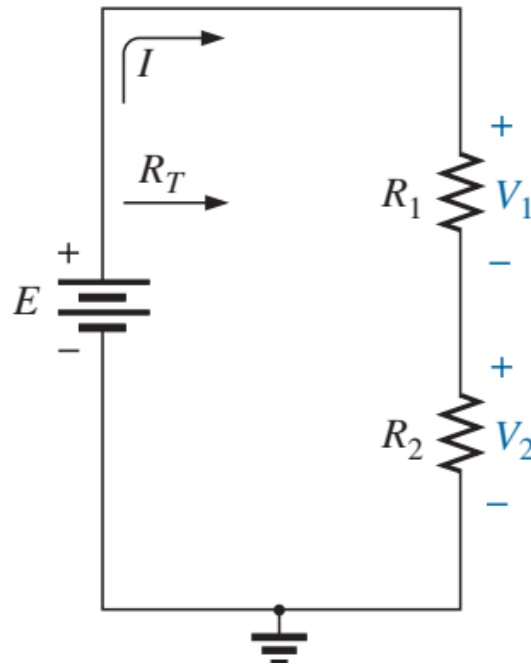
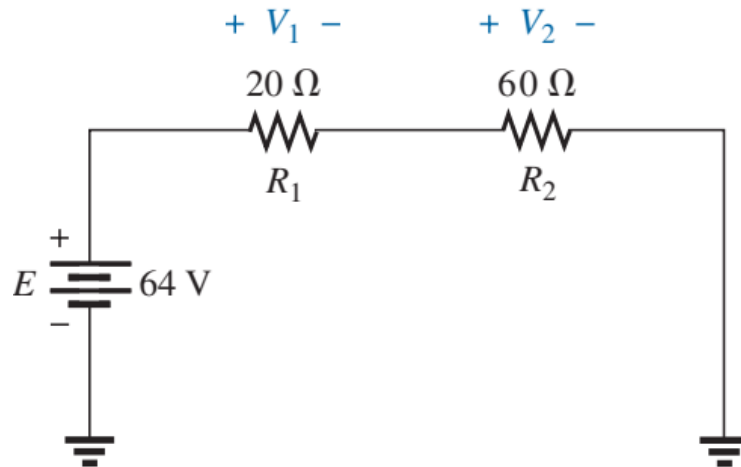


Figure 20: Developing the voltage divider rule.

6. Voltage Division in a Series Circuit (continued...)

Example 10 For the series circuit in fig. 21:

- Without making any calculations, how much larger would you expect the voltage across R_2 to be compared to that across R_1 ?
- Find the voltage V_1 using only the voltage divider rule.
- Using the conclusion of part (a), determine the voltage across R_2 .
- Use the voltage divider rule to determine the voltage across R_2 , and compare your answer to your conclusion in part (c).
- How does the sum of V_1 and V_2 compare to the applied voltage?



6. Voltage Division in a Series Circuit (continued...)

Solutions:

a. Since resistor R_2 is three times R_1 , it is expected that $V_2 = 3V_1$.

$$\text{b. } V_1 = R_1 \frac{E}{R_T} = 20 \, \Omega \left(\frac{64 \, \text{V}}{20 \, \Omega + 60 \, \Omega} \right) = 20 \, \Omega \left(\frac{64 \, \text{V}}{80 \, \Omega} \right) = \mathbf{16 \, \text{V}}$$

$$\text{c. } V_2 = 3V_1 = 3(16 \, \text{V}) = \mathbf{48 \, \text{V}}$$

$$\text{d. } V_2 = R_2 \frac{E}{R_T} = (60 \, \Omega) \left(\frac{64 \, \text{V}}{80 \, \Omega} \right) = \mathbf{48 \, \text{V}}$$

The results are an exact match.

$$\text{e. } E = V_1 + V_2$$

$$64 \, \text{V} = 16 \, \text{V} + 48 \, \text{V} = \mathbf{64 \, \text{V}} \quad (\text{checks})$$

6. Voltage Division in a Series Circuit (continued...)

Homework!

Example 11 For the series circuit in fig. 22, Using VDR, determine voltages V_1 and V_3 .

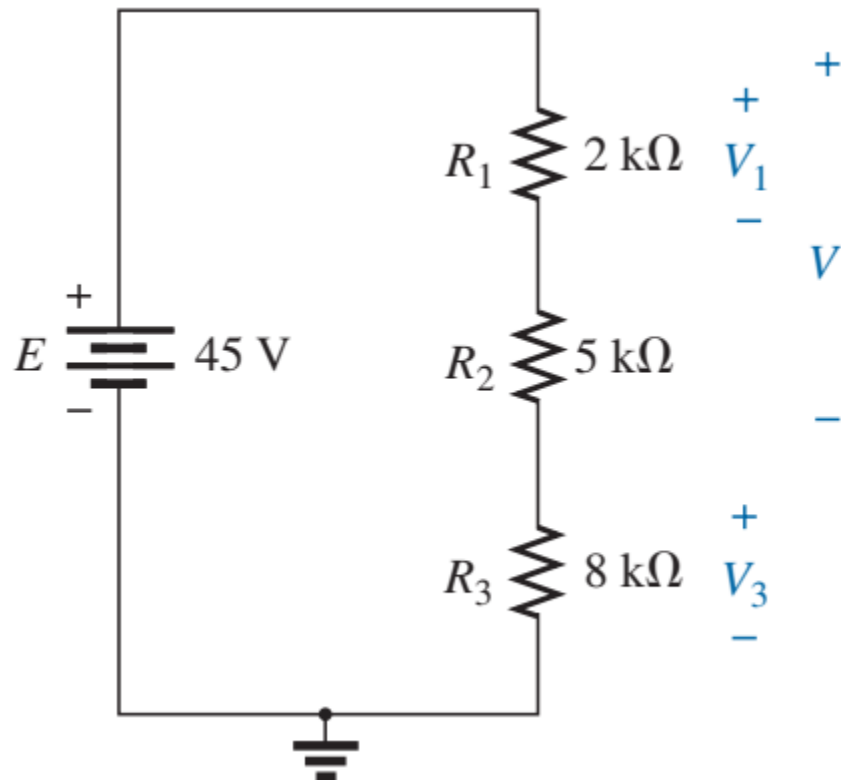


Figure 22: Series circuit to be investigated in Example 11.

6. Voltage Division in a Series Circuit (continued...)

Example 12 Design the voltage divider circuit in Fig. 23 such that the voltage across R_1 will be four times the voltage across R_2 ; that is, $V_{R1} = 4V_{R2}$

Solution: The total resistance is defined by

$$R_T = R_1 + R_2$$

However, if

$$V_{R1} = 4V_{R2}$$

then

$$R_1 = 4R_2$$

so that

$$R_T = R_1 + R_2 = 4R_2 + R_2 = 5R_2$$

Applying Ohm's law, we can determine the total resistance of the circuit:

$$R_T = \frac{E}{I_s} = \frac{20 \text{ V}}{4 \text{ mA}} = 5 \text{ k}\Omega$$

so

$$R_T = 5R_2 = 5 \text{ k}\Omega$$

and

$$R_2 = \frac{5 \text{ k}\Omega}{5} = 1 \text{ k}\Omega$$

Then

$$R_1 = 4R_2 = 4(1 \text{ k}\Omega) = 4 \text{ k}\Omega$$

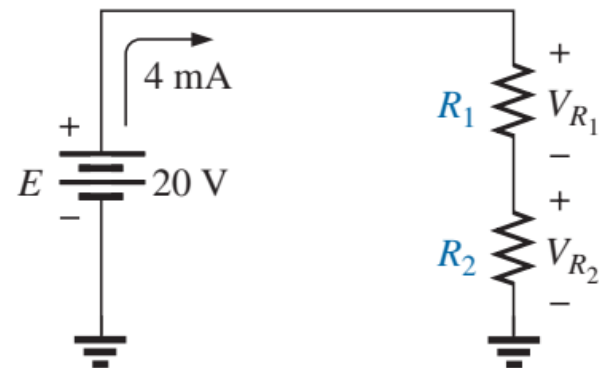


Figure 23: Designing a voltage Divider Circuit.

7. Notation

Voltage Sources and Ground

- Notation plays an increasingly important role in the analysis to follow. It is important, therefore, that we begin to examine the notation used throughout the industry.
- Electrical and electronic systems are grounded for reference and safety purposes. The symbol for the ground connection appears in Fig. 24 with its defined potential level — zero volts. A grounded circuit may appear as shown in Fig. 25(a) and (b). In any case, it is understood that the negative terminal of the battery and the bottom of the resistor R_2 are at ground potential.

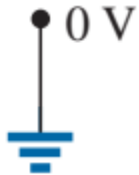
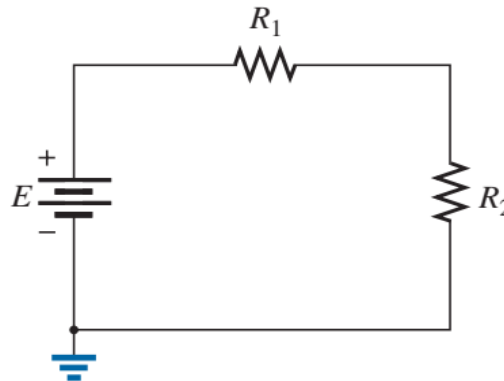
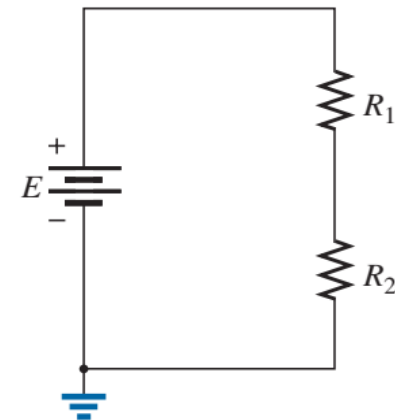


Figure 24: Ground potential.



(a)



(b)

Figure 25: Three ways to sketch the same series dc circuit.

7. Notation (continued...)

- On large schematics where space is at a premium and clarity is important, voltage sources may be indicated as shown in Figs. 26(a) and 27(a) rather than as illustrated in Figs. 26(b) and 27(b).

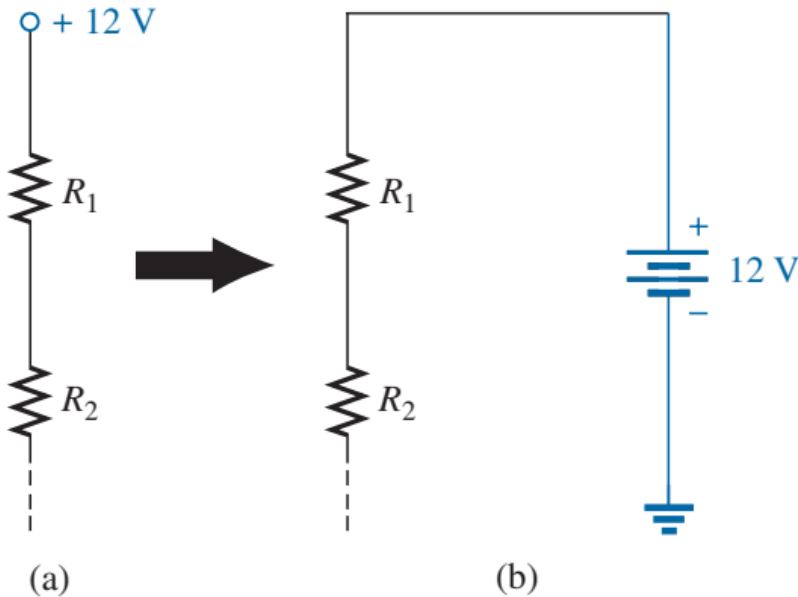


Figure 26: Replacing the special notation for a dc voltage source with the standard symbol.

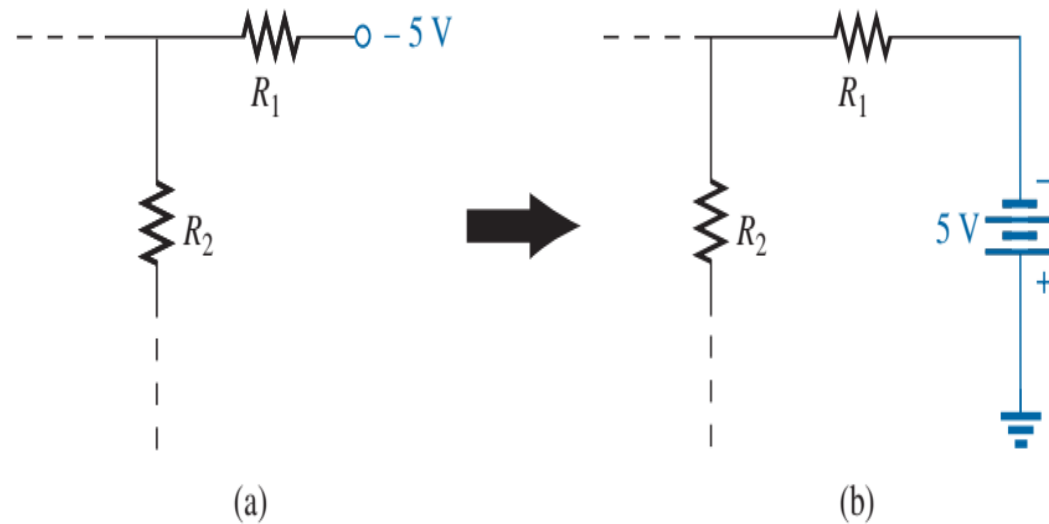


Figure 27: Replacing the notation for a negative dc supply with the standard notation.

7. Notation (continued...)

Double-Subscript Notation

- A double-subscript notation defines the first subscript as the higher potential.
- *The double-subscript notation V_{ab} specifies point a as the higher potential. If this is not the case, a negative sign must be associated with the magnitude of V_{ab} .*
- In other words, the voltage V_{ab} is the voltage at point a with respect to (w.r.t.) point b .

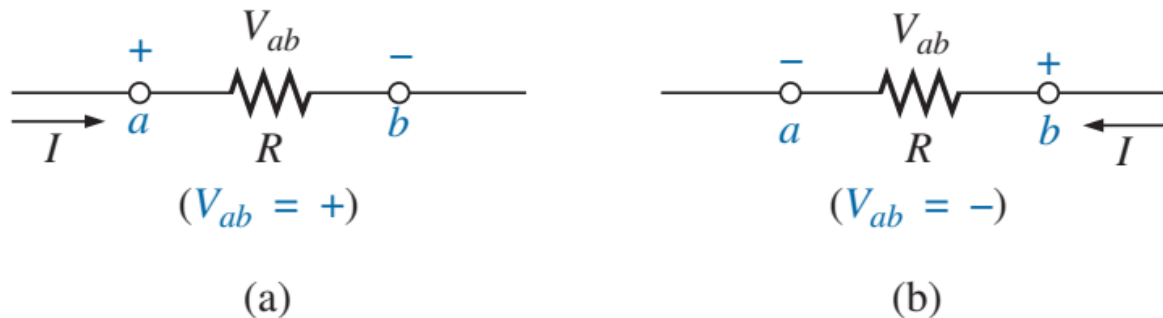
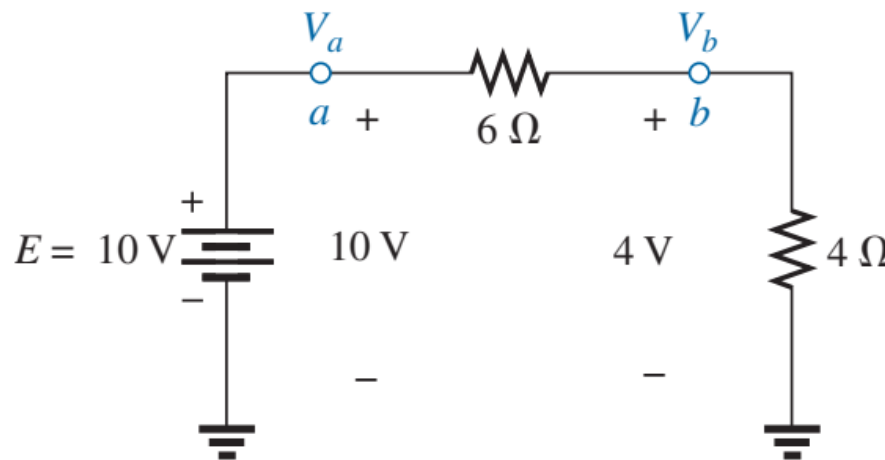


Figure 28: Defining the sign for double-subscript notation.

7. Notation (continued...)

Single-Subscript Notation

- The single-subscript notation V_a specifies the voltage at point **a** with respect to ground (zero volts). If the voltage is less than zero volts, a negative sign must be associated with the magnitude of V_a .
- In Fig. 29, V_a is the voltage from point **a** to ground. In this case, it is obviously 10 V since it is right across the source voltage E . The voltage V_b is the voltage from point **b** to ground. Because it is directly across the 4Ω resistor, $V_b = 4\text{ V}$.



7. Notation (continued...)

General Comments

- A particularly useful relationship can now be established that has extensive applications in the analysis of electronic circuits. For the above notational standards, the following relationship exists:

$$V_{ab} = V_a - V_b$$

- In other words, if the voltage at points *a* and *b* is known with respect to ground, then the voltage V_{ab} can be determined in Fig.29 previous slide, for example.

$$\begin{aligned} V_{ab} &= V_a - V_b = 10 \text{ V} - 4 \text{ V} \\ &= 6 \text{ V} \end{aligned}$$

7. Notation (continued...)

Example 13 For the series circuit in fig. 30, find the voltage V_{ab} .

Solution:

$$\begin{aligned} V_{ab} &= V_a - V_b = 16 \text{ V} - 20 \text{ V} \\ &= \mathbf{-4 \text{ V}} \end{aligned}$$



Figure 30

7. Notation (continued...)

Homework!

Example 14 For the series circuit in fig. 31, find the voltage V_{ab} .

.

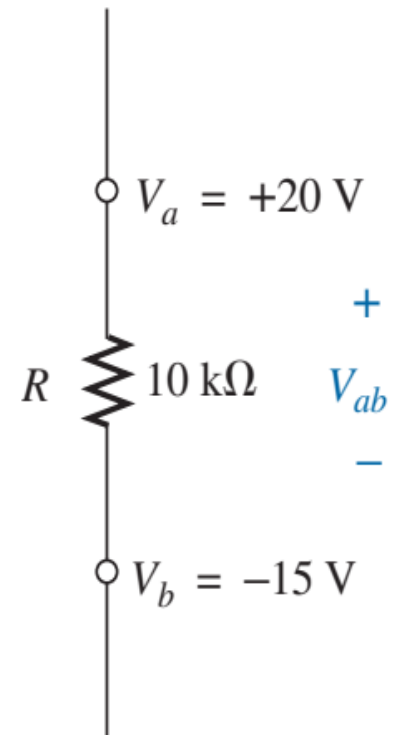


Figure 31:

7. Notation (continued...)

Example 15 For the circuit in fig. 32, find the voltages V_b , V_c , and V_{ac} .

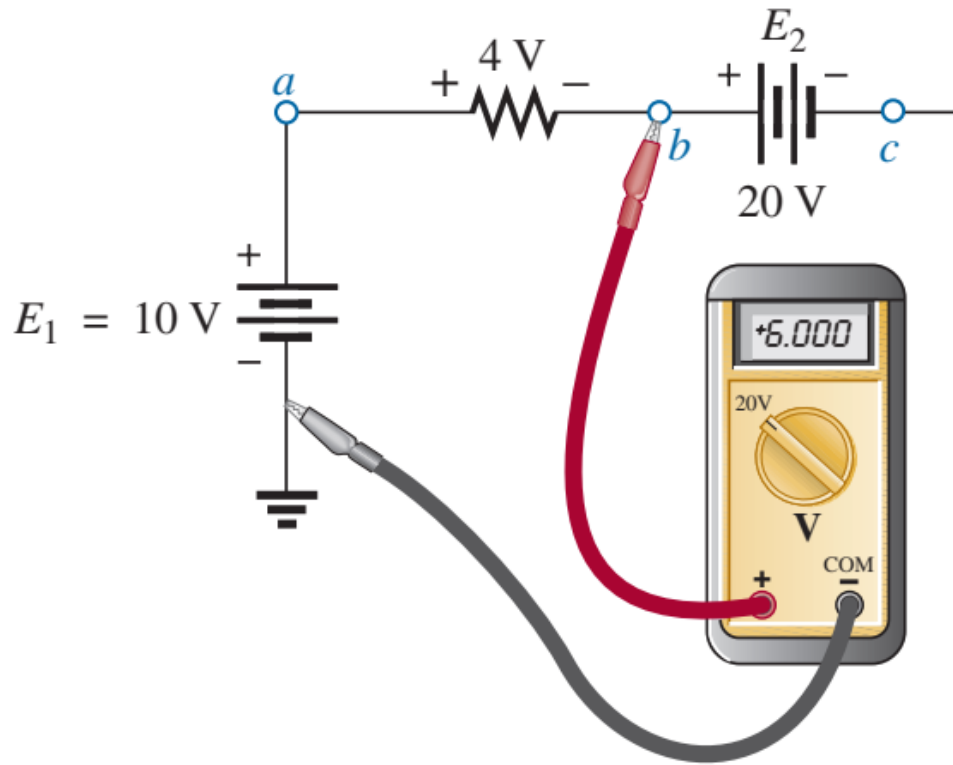


Figure 32

7. Notation (continued...)

Example 15 For the circuit in fig. 32, find the voltages V_b , V_c , and V_{ac} .

Solution: Starting at ground potential (zero volts), we proceed through a rise of 10 V to reach point a and then pass through a drop in potential of 4 V to point b . The result is that the meter reads

$$V_b = +10 \text{ V} - 4 \text{ V} = \mathbf{6 \text{ V}}$$

If we then proceed to point c , there is an additional drop of 20 V, resulting in

$$V_c = V_b - 20 \text{ V} = 6 \text{ V} - 20 \text{ V} = \mathbf{-14 \text{ V}}$$

The voltage V_{ac} can be obtained

$$\begin{aligned} V_{ac} &= V_a - V_c = 10 \text{ V} - (-14 \text{ V}) \\ &= \mathbf{24 \text{ V}} \end{aligned}$$

7. Notation (continued...)

Homework!

Example 16 For the circuit in fig. 33, find the voltages V_b , V_c , and V_{ac} .

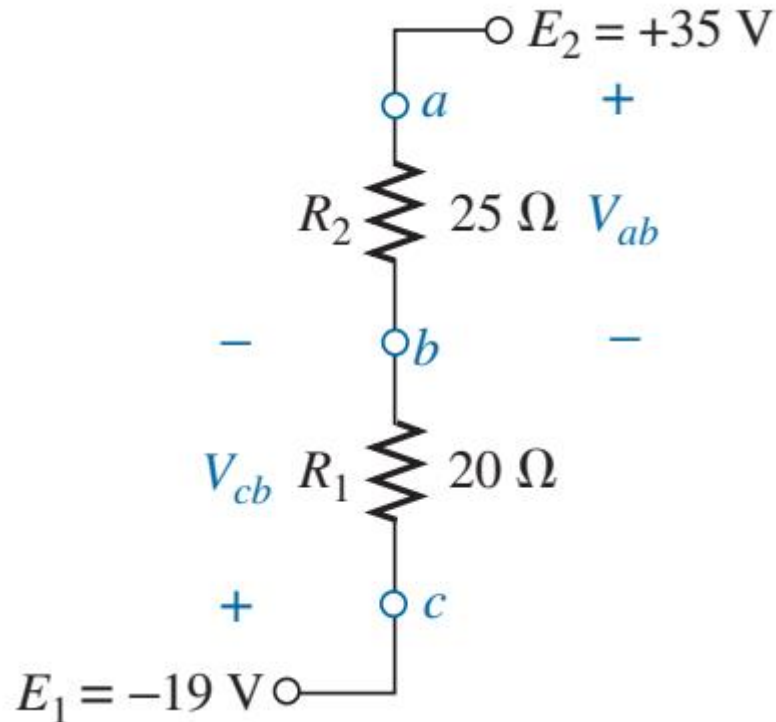


Figure 33:

6. Voltage Division in a Series Circuit (continued...)

Homework!

Example 17 For the circuit in fig. 34, Using VDR, determine voltages V_1 and V_2 .

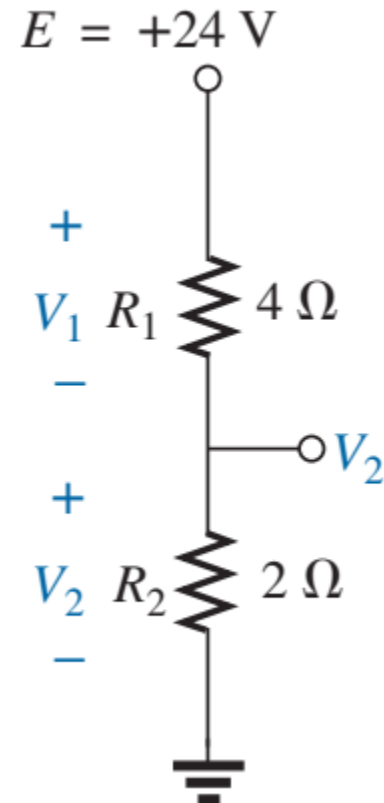


Figure 34:

6. Voltage Division in a Series Circuit (continued...)

Homework!

Example 18 For the network in fig. 35:

- Calculate V_{ab} .
- Determine V_b .
- Calculate V_c .

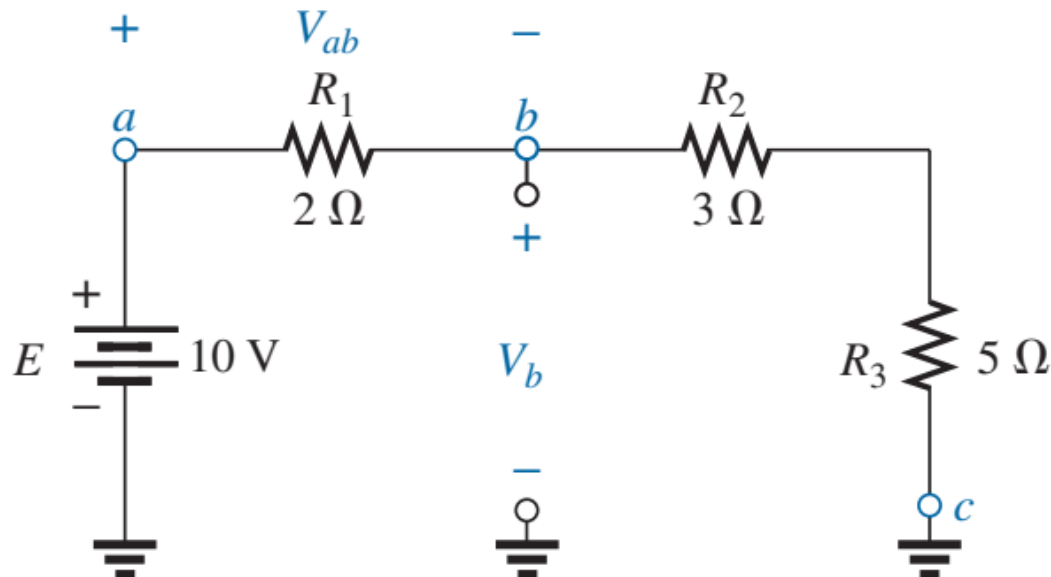


Figure 35:

References: Textbooks

1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12th Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5th Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*